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GRADED PRE-ACTIVITY

1 Regular Expression Task

The regular expression given to us is

$$\text{RE} = 1^*00^*1(0+1)^*.$$

Each component with the operators $*$ and $+$ can be substitued for substrings within the set defined by said operators:

- $1^*:\quad \{\varepsilon, \text{"1"}, \text{"11"}, ...\}$;
- $0^*:\quad \{\varepsilon, \text{"0"}, \text{"00"}, ...\}$;
- $0+1:\quad \{\text{"0"}, \text{"1"}\}$; hence
- $(0+1)^*:\quad \{\varepsilon, \text{"0"}, \text{"1"}, \text{"00"}, \text{"01"}, \text{"10"}, \text{"11"}, \text{"000"}, ...\}$.

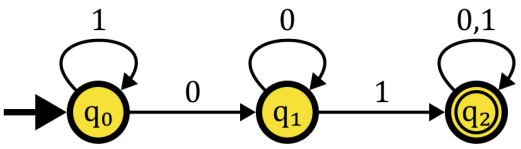
For our example string, we will make the following substitutions:

- 1^* for ε,
- 0^* for "0", and
- $(0+1)^*$ for "10".

And so our example string becomes

$$\begin{aligned}\text{Example String} &= 1^*00^*1(0+1)^* \\ &= (\varepsilon)0(0)1(10) \\ &= 00110.\end{aligned}$$

One possible DFA that can accept this regular expression is given below:



Formally:

$$\begin{aligned}\text{DFA} &= (Q, \Sigma, \delta, q_0, F) \quad \text{such that} \\ Q &= \{q_0, q_1, q_2\}; \\ \Sigma &= \{0, 1\}; \\ q_0 &= q_0; \\ F &= \{q_2\}; \quad \text{and} \\ \delta &\text{ is the transition mapping function defined by below.}\end{aligned}$$

	0	1
$\rightarrow q_0$	q_1	q_0
q_1	q_1	q_2
$*\, q_2$	q_2	q_2

We can prove that our example string, "00110" is accepted by the proposed DFA by walking through the transitions:

1. $\delta(q_0, 0) = q_1$;
2. $\hat{\delta}(q_0, \text{"00"}) = \delta(\delta(q_0, 0), 0) = \delta(q_1, 0) = q_1$;
3. $\hat{\delta}(q_0, \text{"001"}) = \delta\Big(\hat{\delta}(q_0, \text{"00"}), 1\Big) = \delta(q_1, 1) = q_2$;
3. $\hat{\delta}(q_0, \text{"0011"}) = \delta\Big(\hat{\delta}(q_0, \text{"001"}), 1\Big) = \delta(q_2, 1) = q_2$; and finally
3. $\hat{\delta}(q_0, \text{"00110"}) = \delta\Big(\hat{\delta}(q_0, \text{"0011"}), 0\Big) = \delta(q_2, 0) = q_2$;

The final state q_2 was reached from the starting state q_0, hence the example string "00110" is indeed accepted by the DFA that adheres to the regular expression $1^*00^*1(0 + 1)^*$.

2 Machine Code Translation Task

The instructions that must be created using Assembly are as follows:

```
x: int
x = 100
```

One possible program on Windows would look like this:

```
section .text
    global  start
    extern  ExitProcess     ; This is needed for proper exiting.

section .bss
    x       resd 1          ; Memory has to be allocated for the integer x.
                            ; 4 bytes are reserved in this case to align with C.
                            ; This was an arbitary choice.

start:
    mov     dword [x], 100  ; The value 100 is stored within x.
    push    0               ; This status call indicates no errors, like C does.
    call    ExitProcess     ; The program has to be exited properly.
```